

Session 5: Software Part 2

Tech Skills 101: Driver's Ed for the Digital World

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For awareness

Note: You can learn more about by going to <https://www.apple.com/os/> in your browser

- Apple has release new **major** OS versions
 - macOS Tahoe 26.0.1
 - iOS 26.0.1
- If you have not yet upgraded, consider waiting until the 26.1 versions of both are available—which is likely to be as early as the week of October 27
 - The 26.0 versions have some issues that cause “sluggish” behaviors and drain the battery
- When you do upgrade, be prepared to spend time exploring changes — especially explore apps you use / depend on the most

Agenda

- File system basics — including (a lot of) terms
 1. Files and folders are arranged in a hierarchy
 2. Some apps *implicitly* save/read stored data for you and some apps require you to take explicit steps to save/read stored data
- Common scenarios
 - File manager apps
 - Accessing files via an app
- Time permitting: Backing up and cleaning up

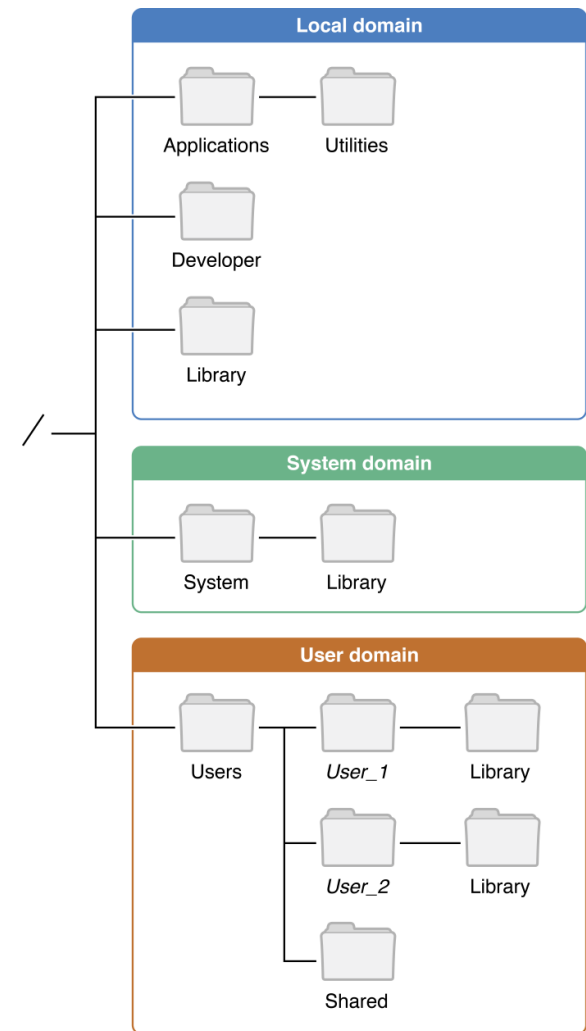
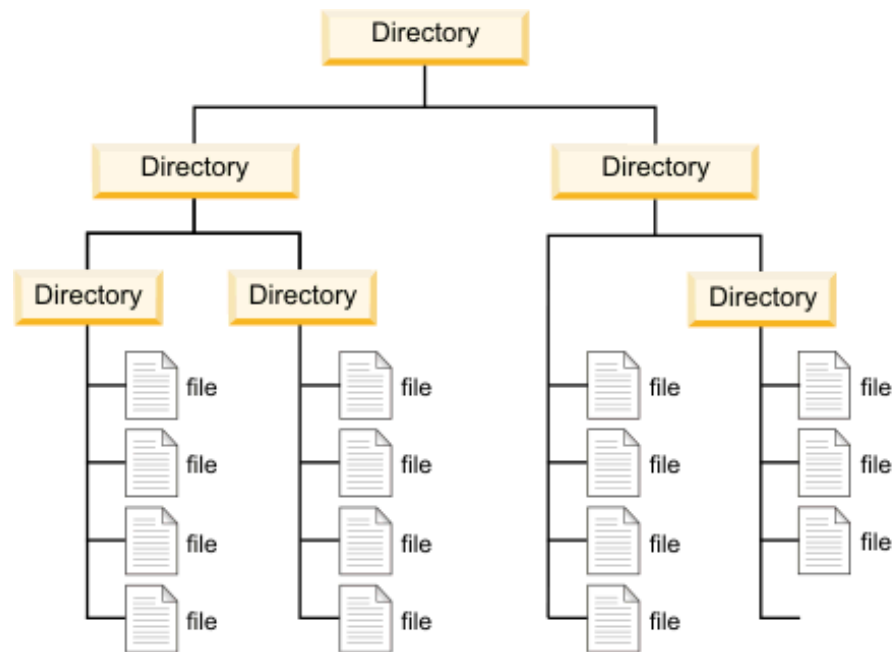
File System Terminology

drive	• A storage device • Short for “disk drive” • Kinds • Internal: the storage device found inside your desktop, laptop, smartphone, and/or tablet • External: a HDD or SSDD that you attach to your desktop/laptop/smartphone/tablet using a physical USB connection • Cloud: storage space “in the cloud” (often Apple’s iCloud, Microsoft OneDrive, or Google Drive)
file	• A collection of digital data stored as a single object on a drive • The type of data stored in a file depends on the file type
folder	• A collection of files and/or other folders • Directory is a synonym for “folder” • Note: Some apps implement a kind of folder to help you organize things in the app itself (e.g., browser bookmark folders); these folder <u>are not</u> file system folders
path	• The location on a drive of a file or folder
alias	• A “shortcut” (also referred to as a “reference”) to a file or folder • An alias looks and acts like a the file or folder it refers to
metadata	• Data that defines/describes the characteristics of other data • E.g., the title, author and publication date of a book are metadata about the book
smart folder	• A folder whose contents using some criteria you specify. • Criteria can include things like age, type, etc

File System Metadata

name	• File and folder names can be up to 255 characters long • File/folder names cannot contain some characters (E.g., “/“ and “\”
type	• Data is arranged in a file in some <i>format</i> • The type (of “kind”) of a file indicates it’s format • A file’s type is indicated by • Its icon • Its file name extension (which may not be shown to you in the UI) • The OS uses a file types to associate files with apps • This is how the OS knows which app to open a file in when you open a file using a file manager
• date/time created • date/time last accessed • date/time last modified	• The date and time the file or folder was created, last opened (i.e., accessed/read), last changed (i.e., modified/updated) • The file system uses this metadata to • To populate your “Recent” smart folder • To remove items in the Trash folder after some period of time
geolocation	• The physical location you were at when the file was created • Geolocation metadata is only included with some types of files • Pictures/videos you take with your device are a primary example

File System Hierarchy



Special Folders

Notes:

- The names used here are “Apple centric”
- Some of these special folders have different names on Windows
- Most of these special folders have not been present on iPhones and iPads—although that appears to be changing!
- Use the green special folders “to your heart’s delight”; leave the pink one alone!

Applications	Contains the code of apps you have installed on your device
Documents	The <i>defacto</i> location for storing files you create using apps on desktops/laptops iPhones and iPads do not include this special folder
Downloads	Location a file is copied to when you download a file from the Internet
Library	• Contain files relating to the operating system and apps you have installed • OS and app settings are stored here • There are three different Library folders: • System Library: contains files specifically related to the operating system • Root Library: contains app related files that affect every user of the desktop/laptop • User Library: contains app related files that affect your desktop/laptop configuration
System	Location of the operating system
Trash	• Always (and only?) find on the macOS doc • Only found on desktops and laptops; not found on iPhones or iPads
Users	• Contains user account “profiles” • Only found on desktops and laptops; not found on iPhones or iPads

File manager apps

A **file manager** or **file browser** is a computer program that provides a user interface to manage files and folders.

from wikipedia.org

On **macOS**, the file manager is called **Finder** and the app icon looks like:



On **iOS**, the file manager is called **Files** and the app icon looks like:



On **Windows**, the file manager is called **File Explorer** and the app icon looks like:



File manager app functions

- **Open** files
- **Search** for files and folders
- **Copy** files and folders
- **Move** files and folders from one folder to another
- **Delete** files and folders
- **Modify** file and folder properties

Demo

- An app that implicitly saves your data to the file system: Contacts
- With Finder
 - Moving around the hierarchy
 - Changing views
 - Creating new folders
 - Moving files and folders
 - Moving a file to your desktop
 - Deleting files and folders
 - The sidebar: Favorites, cloud drives, external drives
 - Mount/eject an external drive
- An app that requires you to explicitly save your data: Pages
 - Look at the File menu
 - Creating a new file
 - Using the same name as an existing file
 - Opening an existing file

Backing up & Cleaning up